SWOT Analysis

Credit-risk modelling frameworks

Framework	SWOT Category	Details
Traditional scorecard	**Strengths**	• High transparency. • Easy to document and challenge. • Familiar to credit-risk practitioners. • Suitable as a benchmark.
	Weaknesses	• May underrepresent nonlinear interactions. • Requires careful binning, transformations, and stability monitoring. • Can be less flexible for complex feature relationships.
	Opportunities	• Combine with monotonic constraints, scorecard development, and calibration. • Use as a challenger model against boosting.
	Threats	• May be outperformed on discrimination by well-governed ensemble models. • Can become outdated when borrower populations or economic conditions change.
Academic machine-learning benchmark	**Strengths**	• Rapid experimentation. • Broad algorithm comparison. • Low licensing cost. • Useful for methodological teaching.
	Weaknesses	• Random splits can overstate future performance. • Preprocessing can leak information. • Accuracy or F1 may be prioritised over PD calibration. • Static thresholds may ignore financial costs. • Explanations may be added after modelling without connecting them to governance.
	Opportunities	• Improve using out-of-time validation, calibration, leakage controls, and decision-cost analysis.
	Threats	• Misleading reported performance. • Poor reproducibility. • Inappropriate deployment or regulatory interpretation.
Commercial AI platform	**Strengths**	• Integrated experiment tracking, deployment, monitoring, and governance. • Potentially faster operationalisation. • Support for model comparison and workflow automation.
	Weaknesses	• Cost and vendor lock-in. • Proprietary components may limit independent auditability. • Outputs still depend on the institution's data, target, and validation design.
	Opportunities	• Used as an implementation environment after a sound research methodology has been established.
	Threats	• Automation can create false confidence. • Governance weaknesses may be hidden behind a polished interface.
Proposed framework	**Strengths**	• Uses 603,589 resolved 36-month loans rather than the earlier small snapshot. • Explicitly defines the target. • Uses chronological train, validation, and test periods. • Separates PD inputs from repayment outcomes. • Stores raw, processed, feature-rule, and preprocessing artifacts. • Compares a borrower/application baseline with a lender-pricing feature set. • Supports calibration, SHAP, and later Expected Loss analysis.
	Weaknesses	• Historical Lending Club data do not represent all lenders or all applicants. • The cohort contains originated loans only. • The target is final observed status, not a fixed-horizon PD. • No full protected-group fairness audit is currently possible. • LGD and EAD are not yet empirically validated. • The current system is a reproducible research pipeline, not a production lending platform.
	Opportunities	• Add calibrated PD models. • Compare Logistic Regression with LightGBM or another ensemble. • Add SHAP explanations. • Estimate or scenario-test LGD and EAD. • Add population-stability and performance-drift monitoring. • Add fairness analysis when appropriate governed data are available.
	Threats	• Label and cohort selection bias. • Vintage and economic drift. • Data-source changes. • Misinterpretation of illustrative fairness or Expected Loss assumptions. • Overclaiming regulatory compliance from explainability alone.

Table 2.2: SWOT Analysis of Industry Credit Models versus the Proposed Framework